GFS SPECIFICATION

DESIGN AND CONSTRUCTION OF STEEL STRUCTURES FOR ONSHORE LNG FACILITIES

GFS IG.34.85.21-Gen.

February 2018

ECCN EAR99

GENERAL FUNCTIONAL STANDARD

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These views are based on the experience acquired during involvement with the design, construction, operation and maintenance of processing units and facilities. Where deemed appropriate GFSs are based on Shell DEPs, international, and industry standards.

The objective is to set the standard for good design and engineering practice to be applied by Shell companies in oil and gas production, oil refining, gas handling, gasification, chemical processing, or any other such facility, and thereby to help achieve maximum technical and economic benefit from standardization.

The information set forth in these publications is provided to Shell companies for their consideration and decision to implement. This is of particular importance where GFSs may not cover every requirement or diversity of condition at each locality.

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3) Contractors/subcontractors and Manufacturers/Suppliers under a contract with users referred to under 1) or 2) which requires that tenders for projects, materials supplied or - generally - work performed on behalf of the said users comply with the relevant standards.

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All administrative queries should be directed to the Technical Standards Team in Shell GSI.

TABLE OF CONTENTS

1.	**INTRODUCTION**	**4**
1.1	SCOPE	4
1.2	DISTRIBUTION, INTENDED USE AND REGULATORY CONSIDERATIONS	4
1.3	DEFINITIONS	4
1.4	CROSS-REFERENCES	6
1.5	SUMMARY OF MAIN CHANGES	6
1.6	COMMENTS ON THIS GFS	6
1.7	DUAL UNITS	6
1.8	NON NORMATIVE TEXT (COMMENTARY)	6
1.9	UPDATE POLICY	7
2.	**DESIGN REQUIREMENT**	**8**
2.1	GENERAL	8
2.2	HIGH/LOW TEMPERATURE DESIGN	9
2.3	CONNECTION DESIGN	9
3.	**MATERIAL REQUIREMENT**	**11**
3.1	STRUCTURAL STEEL (PRIMARY AND SECONDARY STEEL)	11
3.2	BOLT ASSEMBLIES	12
3.3	WELDING CONSUMABLES	12
4.	**MATERIAL HANDLING, SHIPPING AND ERECTION**	**13**
4.1	HANDLING AND SHIPPING	13
4.2	DELIVERY	13
4.3	MARKING SYSTEM	14
5.	**CORROSION PROTECTION**	**15**
5.1	GENERAL	15
5.2	PAINTING	15
5.3	HOT-DIP GALVANISATION	15
6.	**CONSTRUCTION REQUIREMENT**	**17**
6.1	GENERAL	17
6.2	FABRICATION AND FABRICATION DRAWINGS	17
6.3	ERECTION	19
6.4	CORRECTION OF ERRORS	20
7.	**CONSTRUCTION HSSE REQUIREMENTS**	**21**
7.1	SAFETY	21
7.2	ERECTION/ASSEMBLY LIFT PLAN	21
7.3	HOLE COVERS AND OPENING BARRIERS	22
8.	**QUALITY CONTROL**	**23**
8.1	GENERAL	23
8.2	WELDING AND WELD INSPECTION	23
8.3	DOCUMENTATION	24
9.	**REFERENCES**	**25**

1. INTRODUCTION

1.1 SCOPE

This new GFS specifies requirements and gives recommendations for the design, fabrication and erection of steel structures for projects for onshore LNG facilities. This GFS applies to stick-built steel structures. It specifies the industry codes, standards and additional technical requirements (including safety related requirements) that comprise the minimum technical requirements to be applied to the Onshore LNG facilities that are covered by the scope of this GFS.

This GFS is classified as a TIER 2 document. See (1.3) for definitions of TIER 1/TIER 2.

1.2 DISTRIBUTION, INTENDED USE AND REGULATORY CONSIDERATIONS

Unless otherwise authorised by Shell GSI, the distribution of this GFS is confined to Shell companies and, where necessary, to Contractors nominated by them. Any authorised access to a GFS does not for that reason constitute an authorisation to any documents, data or information to which the GFS may refer.

This GFS is intended for use in liquefied natural gas (LNG) production facilities and associated supply/marketing. This GFS may also be applied in other similar facilities.

When GFSs are applied, a Management of Change (MOC) process shall be implemented; this is of particular importance when existing facilities are to be modified.

If national and/or local regulations exist in which some of the requirements could be more stringent than in this GFS, the Contractor shall determine by careful scrutiny which of the requirements are the more stringent and, which combination of requirements will be acceptable with regards to safety, environmental, economic and legal aspects. In all cases, the Contractor shall inform the Principal of any deviation from the requirements of GFS which is considered to be necessary in order to comply with national and/or local regulations. The Principal may then negotiate with the Authorities concerned the objective being to obtain agreement to follow this GFS as closely as possible.

1.3 DEFINITIONS

1.3.1 General definitions

The **Contractor** is the party that carries out all or part of the design, engineering, procurement, construction, commissioning or management of a project or operation of a facility. The Principal may undertake all or part of the duties of the Contractor.

The **Manufacturer/Supplier** is the party that manufactures or supplies equipment and services to perform the duties specified by the Contractor.

The **Inspector** is the authorized representative with authority to act in the interest of and on behalf of the Contractor in all quality assurance matters.

The **Erector** is the party that is responsible for the erection of the structural and miscellaneous steel. Unless otherwise noted, the term Erector shall also apply to the Erector's subcontractor(s) and/or vendor(s).

The **Fabricator** is the party responsible for furnishing fabricated structural and miscellaneous steel. The Fabricator typically obtains materials from the mill, develops the shop drawings to fabricate all steel members, and develops the erection drawings.

The **Principal** is the party that initiates the project and ultimately pays for it. The Principal may also include an agent or consultant authorised to act for, and on behalf of, the Principal.

TIER 1 packaged equipment solutions are kept evergreen and have a revision cycle.

TIER 2 packaged equipment solutions do not have a revision cycle and updated when needed.

The word **shall** indicates a requirement.

The word **should** indicates a recommendation.

The word **may** indicates a permitted option.

1.3.2 Specific definitions

Term	Definition
Corrosive Environment	An environment that has one or more of the following elements including moist air, hot air, hot water, acids and salt water.

1.3.3 Abbreviations

Term	Definition
AFC	Approved for Construction
AISC	American Institute of Steel Construction
ANSI	American National Standards Institute
ASTM	American Society for Testing and Materials
AWS	American Welding Society
CVN	Charpy V-Notch
DTI	Direct Tension Indicator
GFS	General Functional Standard
HSS	Hollow Structural Standard
IG	Integrated Gas
ITP	Inspection and Test Plan
JSA	Job Safety Analysis
LNG	Liquefied Natural Gas
LRFD	Load Resistance Factor Design
NAAM	National Association Of Architectural Metal Manufacturers
RA	Risk Analysis
SI	International System
TAC	Test Acceptance Criteria
TPQ	Test Procedure Qualifications
TS	Task Statement
USC	US Customary Measurement Units

Term	**Definition**
WP	Welding Procedure
WPS	Welding Procedure Specification
WQR	Welder Qualification Records

1.4 CROSS-REFERENCES

Where cross-references to other parts of this GFS are made, the referenced section or clause number is shown in brackets ( ). Other documents referenced by this GFS are listed in (9).

1.5 SUMMARY OF MAIN CHANGES

This is a new GFS.

1.6 COMMENTS ON THIS GFS

Comments on this GFS may be submitted to the Administrator using the same methods as for submitting comments on DEPs:

Shell DEPs Online (Users with access to Shell DEPs Online)	Enter the Shell DEPs Online system at https://www.shelldeps.com Select a GFS and then go to the details screen for that GFS. Click on the “Give feedback” link, fill in the online form and submit.
DEP Feedback System (Users with access to Shell Wide Web)	Enter comments directly in the DEP Feedback System which is accessible from the Technical Standards Portal http://sww.shell.com/standards. Select “Submit DEP Feedback”, fill in the online form and submit.
DEP Standard Form (other users)	Use Standard Form PES 00.00.05.82-Gen. to record feedback and email the form to the Administrator at standards@shell.com.

Feedback that has been registered in the Feedback System by using one of the above options will be reviewed by the GFS Custodian for potential improvements to the GFS.

1.7 DUAL UNITS

This GFS contains both the International System (SI) units, as well as the corresponding US Customary (USC) units, which are given following the SI units in brackets. When agreed by the Principal, the indicated USC values/units may be used.

1.8 NON NORMATIVE TEXT (COMMENTARY)

Text shown in italic style in this GFS indicates text that is non-normative and is provided as explanation or background information only.

Non-normative text is normally indented slightly to the right of the relevant GFS clause.

1.9 UPDATE POLICY

This GFS will be updated when applied in a project in accordance with the framework for TIER 2 Packaged Equipment Solutions.

2. DESIGN REQUIREMENT

2.1 GENERAL

1. The design of steel structures shall be in accordance with the following standards:
    a. ANSI/AISC 360 - Specification for Structural Steel Buildings;
    b. ANSI/AISC 303 - Code of Standard Practice for Steel Buildings and Bridges.

        Alternative standards can be considered if essential for compliance with local regulations.
2. Design methodology shall be Load and Resistance Factor Design (LRFD) in accordance with AISC 360.
3. Loading criteria and combinations shall be in accordance with GFS IG.34.85.02-Gen.
4. Deflections shall be verified using the serviceability combinations documented in GFS IG.34.85.02-Gen.
5. Foundations shall be designed in accordance with GFS IG.34.85.25-Gen.
6. For overhead runway beams with underhung cranes, the bottom flanges shall be checked separately for distortion and reinforced if required.
    a. For lower capacity cranes, joists with inside tapered flanges are preferred, with a minimum thickness of 6 mm (1/4 in).
7. The minimum flange width shall be as follows:
    a. 80 mm (3 ¼ in) shall be provided for purlins supporting lightweight concrete slabs;
    b. 50 mm (2 in) shall be provided for supporting roof sheeting and wall cladding.
8. Gusset plates shall not be thinner than the members to be connected with a minimum thickness of 10 mm (3/8 in).
9. Beam end plates shall not be thinner than the flange thickness of that beam with a minimum thickness of 10 mm (3/8 in).
10. Unless otherwise noted on the drawings, all structural member bolted connections shall use high strength bolts (3.3.1) with a minimum 20 mm (3/4 in) diameter.
11. 16 mm (5/8 in) diameter bolts or smaller may be used for ladders, stair treads, purlings, girts, door frames and handrail connections.
12. Maximum bolt holes shall be in accordance with AISC based on nominal bolt diameter.
13. Unless otherwise noted on the design drawings, the following shall be used as a minimum for connection design:
    a. moment connections shall be designed to develop the full factored bending strength of the weaker connected member;
    b. shear connections shall be designed to resist the factored reaction from the uniform beam load that would create a bending moment equal to the bending capacity of the laterally supported section;
    c. tensile and compressive connections shall be designed to resist at least half of the factored tensile or compressive capacity of the gross cross section of the member.

14. Beam to column connections shall be sized for the most severe load combination of the following:
    a. beam end reaction;
    b. vertical components for the bracing force;
    c. horizontal components of the bracing force;
    d. the axial force transmitted through the column to the beam.

2.2 HIGH/LOW TEMPERATURE DESIGN

1. For structures exposed to “High Temperatures”:
    a. the allowable design stresses of structural steel elements that are continuously exposed to high temperatures above 260 °C (500 °F) shall be reduced in proportion to reductions in yield strength of the steel;
    b. detailing of joints shall account for fatigue cycles and stresses caused by the combined effects of loads and large temperature variations;
    c. corrosion protection suitable for hot climates shall be provided.
2. For structures exposed to “Low Temperatures,” i.e., when the design lowest ambient mean temperature is below 0 °C (32 °F):
    a. steel selected for cold climate application shall satisfy fracture toughness requirement (i.e., minimum CVN impact energy value) based on steel grade and thickness of steel member.

       Reference to appropriate national codes and specific consideration of applicable temperatures. (This can include consideration of fabrication sites/stowage if prefabrication/modularisation construction methods are envisaged).
    b. detailing of joints shall account for fatigue cycles and stresses caused by the combined effects of snow and wind loads and large temperature variations;
    c. corrosion protection suitable for cold climates shall be provided.

2.3 CONNECTION DESIGN

1. The Contractor shall be responsible for the structural design adequacy of all connections.
    a. Structural design adequacy shall be in accordance with AISC.
2. Seismic design of connections shall be in accordance with AISC 341–Seismic Provisions for Structural Steel Buildings and AISC 358–Prequalified Connections for Special and Intermediate Steel Moment Frames for Seismic Applications.
3. Joints and connections shall be designed such that they are accessible for inspection, cleaning and painting.
4. A minimum of two bolts shall be used in a framed-beam and truss connection.
5. Cross bracing connections shall be bolted at intersections with a minimum of:
    a. four bolts for structural tee bracing;
    b. two bolts for angle bracing.
6. Tee bracing shall be connected at the flange.
7. Bolted connections exposed to dynamic loads and vibrations (e.g., supporting machinery, crane beams, flare stacks and masts) shall be provided with washers and lock nuts.
8. Shop connections may be welded/bolted.

9. Field connections shall be bolted unless otherwise shown on the design drawings.
10. Hardened washers shall be used to provide adequate bearing area and to prevent damage to surface protection when bolts are tightened.
11. Where the surfaces of bolted parts, which are in contact with the bolt head or nut, have a slope of more than 1:20 with respect to a plane normal to the bolt axis (e.g., channels and/or I-beams), smooth tapered washers shall be used.
12. Column splices shall be field-bolted.
13. Welded connection design shall be in accordance with ANSI/AWS D1.1.
14. A minimum fillet weld leg size of 5 mm (3/16 in) shall be used for structural welds.
15. A minimum seal weld leg size of 3 mm (1/8 in) fillet welds shall be used.

3. MATERIAL REQUIREMENT

3.1 STRUCTURAL STEEL (PRIMARY AND SECONDARY STEEL)

1. Structural steel shall meet requirements in accordance with Table 3.1.
   a. Alternative materials shall require approval by the Principal.
2. Where alternative materials are selected, weldability and weldment mechanical properties shall be compatible.
3. The procured material shall be 'new' material.
   a. Use of re-used/re-cycled material shall not be permitted.

Table 3.1 Structural Steel

Section/member type	Standard	Remarks
Structural steel shapes	ASTM A992/A992M	Minimum yield strength 350 MPa (50 ksi)
Rolled steel joists/ miscellaneous structural shapes	ASTM A992/A992M, ASTM A36/A36M	
Channels, angles, plates, bars	ASTM A36/A36M	
Banding and toe plate banding	ASTM A36/A36M or ASTM A1011/A1011M	
Steel pipe as structural members (hollow round shapes with thicker wall thickness than round tubing).	ASTM A53/A53M Type E (welded) or S (seamless), Grade B or ASTM A106/A106M Grade B	Structural pipe to be seamless, or mill welded. Spiral-welded pipe is not permitted.
Structural tubing (hollow sections in round, square and rectangular shapes)	ASTM A501 or ASTM A500 Grade B	
Checkered floor plate	ASTM A786/A786M Pattern 4 or Pattern 5; ASTM A36/A36M	
Welded steel bar grating and grating stair treads	ASTM A1011/A1011M	Galvanised in accordance with ASTM A123 and NAAMM MBG 531.
Wall mounted handrail carbon steel, standard weight (round structural tubing or pipe)	ASTM A500 Grade B or ASTM A53 Grade B	48 mm (2 in) outside diameter

3.2 BOLT ASSEMBLIES

3.2.1 High strength bolt assemblies

1. Bolts shall be in accordance with ASTM A325 Type-1 or ASTM A490/A490M.
2. Heavy hex nut and threads shall be in accordance with ASTM A563, Grade DH.
3. Washers shall be in accordance with ASTM F436/F436M.
4. Direct Tension Indicator (DTI) washers shall be of a compressible washer type.
5. DTI washers shall be galvanised in accordance with the DTI Manufacturer specifications and ASTM F959.
6. High-strength bolted connections shall be a bearing type with threads included in the shear plane.
    a. Slip-critical type and friction-grip shall be used when noted in the drawings.
    b. Slip-critical connections shall not be used for sustained loads.

3.2.2 Standard bolt assemblies

1. Standard bolts shall be in accordance with ASTM A307, Grade “A” heavy hex.
2. Heavy hex nuts and threads shall be as per ASTM A563, Grade “A”, “C” and “D”.
3. Washers shall be in accordance with ASTM F436/F436M.

3.3 WELDING CONSUMABLES

1. Welding electrodes shall be in accordance with ANSI/AWS D1.1 (Low hydrogen with an electrode strength of 400 MPa (58 ksi) minimum yield strength and 480 MPa (70 ksi) minimum tensile strength).
2. The minimum requirements for welding material, such as yield strength and tensile strength shall be the same as for the parent material.

4. MATERIAL HANDLING, SHIPPING AND ERECTION

4.1 HANDLING AND SHIPPING

1. Delivery of steel shall be made in the order needed for erection.
2. Bolts, washers and nuts shall be packaged and delivered in rigid (not cardboard), weatherproof containers.
3. Railcars and trucks shall be loaded to optimize the sequence of construction.
4. The Contractor and/or his sub-contractor (i.e., the Manufacturer/Supplier or Erector) shall ensure the following protection:
    a. all steel and steel coatings are protected from damage caused by handling, storage or shipping before delivery to the project site;
    b. protection is provided for the threads on sag rods and any other threaded components to prevent damage during shipping and handling.

4.2 DELIVERY

1. The Contractor and/or his sub-contractor (i.e., the Manufacturer/Supplier or Erector) shall be responsible for delivering all materials and documentation to the job site in good condition.
2. Materials and documentation shall be inspected immediately upon receipt by the Contractor to ensure the following:
    a. all items included in the bill of materials have been supplied;
    b. all documentation has been received;
    c. material is checked for any damage.
3. If parts, components or documentation is missing, or items are found to be damaged or otherwise defective, the occurrence shall be immediately reported in writing.
4. Fabricated structural steel shall be delivered in a sequence that permits efficient and economical fabrication and erection, and in a way, that is consistent with requirements in the contract documents.
5. Materials shall be unloaded, stored and handled in a manner that prevents distortion, deterioration, damage or staining.
6. Materials shall be kept free of dirt, grease and other foreign matter.
7. Anchor rods, washers, nuts and other anchorage or grillage materials that are to be built into concrete or masonry shall be shipped so that they are available when needed.

4.3 MARKING SYSTEM

1. Markings shall be in accordance with the erection plans, shop drawings, ladder drawings, floor drawings and/or handrail drawings (as applicable).
2. Members shall be marked with a low-stress die stamp at the appropriate stage prior to surface preparation.
3. Painted identification marks shall be applied to all completed members prior to shipping.
    a. Stencils shall be used, with letters and numerals having a minimum height approved by the Contractor.
    b. Paint materials shall be of a mix and colour compatible with the finish coat paint.
    c. Application of paint materials shall have sufficient contact for readability.
    d. All markings shall be legible and clearly visible.
    e. For miscellaneous non-structural items where sufficient marking area is not available, a corrosion-resistant metal tag indicating the item mark shall be securely attached to each individual piece by stainless steel wire or resistant coating.
4. In addition to the identification marking mentioned in (4.3, Item 3), a label printed with the structure identification code and colour marking shall be attached to every member after painting/galvanising.
5. Each bolt component shall be clearly marked with the bolt manufacturer’s identification.
6. The Contractor and/or his sub-contractor (i.e., the Manufacturer/Supplier or Erector) shall colour code, die punch or otherwise mark the ends of torqued bolts indicating that the bolts have been properly tensioned and are ready for inspection. “Squirter Type” DTI washers are preferred to ease inspection.
7. The colour marking shall use a combination of colours and figures/shapes.
    a. Usable colours are red, white (blank), blue, green, yellow, orange, pink, purple and black.
8. The Contractor may choose from the following figures/shapes:
    a. square, 30 mm x 30 mm (1 ¼ in x 1 ¼ in);
    b. triangular, 30 mm (1 ¼ in) width;
    c. rectangular, 15 mm x 30 mm (5/8 in x 1 ¼ in);
    d. circular, 30 mm (1 ¼ in) diameter.
9. Colour marking shall be as specified in the engineering drawings and project specifications.
    a. Figure and colour markings shall be clearly shown on the erection plan drawings as information to the Erector.

5. CORROSION PROTECTION

5.1 GENERAL

1. Special protective coating measures and/or material thickness allowance shall be used for structural elements exposed to a corrosive environment.

5.2 PAINTING

1. Surface preparation and painting shall be in accordance with GFS IG.31.87.77-Gen.

5.3 HOT-DIP GALVANISATION

5.3.1 General

1. Hot-dip galvanising shall be in accordance with the following:
    a. ASTM A123 - Standard Specification for Zinc (Hot-Dip Galvanised) Coatings on Iron and Steel Products;
    b. ASTM A153 - Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware.
2. Where hot-dip galvanising is specified, all joints and connections of hot-dip galvanised structures shall be provided with hot-dip galvanised bolts, nuts and washers.
3. Cutting, drilling and welding shall be performed before galvanising.
4. Weld slag shall be removed before galvanising.
5. The Contractor and/or his sub-contractor (i.e., the Manufacturer/Supplier or the Erector) shall safeguard against embrittlement and warpage during the galvanising process.
6. Structural steelwork, platform flooring and hand railing located above austenitic stainless steel, duplex stainless steel (i.e., piping or similar), nickel alloy or 9 % nickel steel piping or equipment, shall not be galvanised or coated with zinc-containing paint unless protection is provided for stainless steel elements.
    a. These precautions shall be taken to prevent zinc contamination of the piping or equipment when hot-working on the galvanised material or applying the zinc-containing paint.
    b. A suitable measure is the application of temporary shielding such as fire blankets. (For components that are insulated, the cladding is considered to be sufficient protection).
7. Nuts shall be tapped oversize.
    a. Nut threads shall be re-tapped after hot-dip galvanizing to provide a proper fit.
8. The Principal shall reserve the right to reject all galvanised steel based on the quality of galvanisation as per project specifications.
    a. Galvanised steel that has been rejected may be stripped, re-galvanised and resubmitted for inspection at the Contractor and/or the sub-contractor (i.e., the Manufacturer/Supplier or Erector) expense.

5.3.2 Repair of Galvanised Steel

1. Damage to galvanizing shall be repaired as follows:
    a. prior to the repair of damaged galvanised coating, the exposed substrate metal shall be cleaned to bright metal and be free of all visual rust, oil or grease.
    b. any non-adhering galvanising shall be removed to the extent that the surrounding galvanising is integral and adherent.
2. When surface defects exceed 2 % of a member's area, the defects shall be repaired by re-dipping the member in the zinc bath.
3. Cold repair using an organic zinc-rich coating shall only be allowed if the total damaged area is less than 1% of the total coated area of the member being repaired and no single repair is greater than 1300 mm^2 (2 in^2) or 300 mm (12 in) long.
4. The dry film thickness shall be 0.05 (0.002 in) – 0.08 mm (0.005 in) and contain a minimum of 65% zinc dust by weight.
5. Hot repairs shall be made in the shop if any of the following conditions exist:
    a. the total damaged area is greater than 1%, but less than 2 %, of the total coated area of the member being repaired;
    b. any single repair is at least 1300 mm^2 (2 in^2) in the area;
    c. any single repair is 300 mm (12 in) long or more.
6. Hot repairs shall be made using a zinc alloy rod or powder manufactured for the repair of galvanised steel.
7. Flux or heavy ash shall be removed by brushing, grinding or filing as required.
8. Excessive warpage shall be corrected by press straightening. (when possible).
    a. The application of localised heating to straighten excessive warpage shall not be permitted.
9. If galvanised tension control bolts are used, all bare steel surfaces (i.e., bolt ends) shall have the galvanised coating repaired.
10. Erection damage to hot-dip galvanised or to shop-applied paint coatings shall be repaired in accordance with the project specifications.
11. Damage to the galvanisation shall be not be painted.

6. CONSTRUCTION REQUIREMENT

6.1 GENERAL

1. Steel structures and their connections shall be constructed in accordance with AISC.
2. Certified Mill Test Reports for structural material including bolts and welding electrodes shall be provided for review by the Contractor.
3. Testing shall be in accordance with applicable codes and standards as specified in the project quality requirements included in the contract.
4. Permitted products and materials shall be new, of top quality and free from defects, loose scale, slag and rust.
5. Bolting material, including washers and nuts, required for erection (plus spares) shall be furnished and delivered with the first shipment of each steel release package.
6. Pockets or depressions, which may collect and hold water, shall be provided with drain holes to protect from corrosion due to standing water.
7. The Principal approval shall be required prior to the start of fabrication for splices in structural steel other than those shown on the engineering drawings.
8. Steel with passive fire protection shall be detailed to provide weather protection in accordance with the project specifications and local regulations.
9. The connection of hand railing, stairs, ladders and bracing (for angle only) fabricated on fire-proofed beam/columns shall be located outside the fireproofing.
10. Hand railing shall be shop fabricated.

6.2 FABRICATION AND FABRICATION DRAWINGS

1. Fabrication drawings shall be reviewed and approved by the Contractor.
2. Fabrication drawings shall take into account the required measures for transport of shop-assembled parts to the construction site.
3. Deviations from the design drawings, shop splices, substitutions of member sizes or changes in details or dimensions, shall not be permitted without written authorisation from the Contractor.
4. Beams, except cantilevers, shall be fabricated with natural mill camber in the up position.
5. A positive camber of 10 mm (3/8 in) for every 5 m (16 ft) span shall be applied if the total span of a structure or frame exceeds 20 m (65 ft).
6. Re-entrant corners shall be shaped notched-free to a radius.
7. Prior to surface preparation, the Manufacturer/Supplier/Fabricator shall remove all sharp corners, burrs (including bolt hole burrs), weld spatter, slag, weld flux, loose mill scale and other foreign matter.
8. A minimum of four fasteners per grating panel shall be used at each support for gratings.
9. Square grating panels shall not be fabricated to avoid any installation errors.
10. Checkered plate shall have 12 mm (1/2 in) diameter drain holes provided for each 2 m^2 (22 ft^2) of area, with a minimum of one hole per panel.
11. Each piece of mill material shall be legibly marked with the heat number, size of section, length and mill identification marks as well as the fabrication mill order number.

12. Bolted connections design and construction shall account for the following:
    a. burning and/or flame cutting of holes shall not be allowed;
    b. crane rail joints shall be staggered in arrangement;
    c. rail splices and crane girder splices shall not coincide with each other;
    d. for slotted holes, the long direction of the slot shall be perpendicular to the load direction, unless otherwise noted in the drawings;
    e. hardened washers shall be provided under all bolt heads and nuts adjacent to any ply with slotted holes;
        i. for standard holes, a minimum of one hardened washer shall be supplied with each bolt;
    f. when two structural members on opposite sides of a column web, or a beam web over a column, are connected sharing common connection holes, the Fabricator shall provide a means of supporting one member while erecting the other member;
    g. unless the means of support is indicated in the drawings, one of the following shall be provided:
        i. one additional row of bolts in the member shall be erected first;
        ii. an erection seat, sized to support the dead weight for the member shall be erected first, unless additional loading is indicated.
    h. temporary bolts used for erection purposes shall be painted in bright fluorescent orange, so as to be spotted from grade;
    i. mechanically galvanised bolts and nuts shall not be intermixed with hot-dip galvanised nuts and bolts.
13. Practical length of bracing under tension (as specified in design drawings) shall be shorter than theoretical to provide nominal pretension (to be confirmed during detail design) in accordance with the Contractor workmanship.
    a. This approach shall be applied with care as the magnitude of load is uncontrolled and will be additive to the externally applied loads.
14. Welds to connection plates embedded in concrete shall be deposited in a sequence that limits the distortion of the embedment to 3 mm (1/8 in).
15. All ragged edges, welds, protruding bolts or other fasteners that might cause injury to personnel shall be avoided, or if required, ground smooth.
16. All hollow sections exposed to the outdoor environment shall be sealed.
    a. The ends shall be sealed (welded) with 6 mm (1/4 in) thick cap plates.
    b. Galvanised closed HSS assemblies shall be provided with vent/drain holes to avoid rapid pressure changes during galvanising process.
17. Mechanical operating devices such as hinges, hasps, hold-opens and like items shall operate smoothly and efficiently.
18. Ladder cages and safety gates shall be shop assembled on to the ladders.
19. Platforms, stairways and handrails shall be shop assembled in the largest units suitable for handling and shipping.

6.3 ERECTION

1. Tools, equipment and ropes used for erection shall be certified for the purpose intended.
2. Prior to erection activities, method statements shall be submitted by the implementing Contractor.
3. Before commencing work, the Contractor and/or his sub-contractor (i.e., the Manufacturer/Supplier or Erector) shall survey and check foundations and other connection points to confirm their location, orientation, elevation and condition.
    a. If there is incomplete or unacceptable work that affects the erection, it shall be immediately reported to the Contractor in writing and recorded in the daily erection record.
4. Before starting to dress the elevation of a structure with secondary members to main members, plumb main columns and main beams between the splices.
    a. As soon as the structure is properly aligned, plumbed, levelled and braced, erection with permanent bolts and all connections shall be completed.
5. The Contractor and/or his sub-contractor (i.e., the Manufacturer/Supplier or Erector) shall be responsible for the following:
    a. setting out the works and accurately positioning, plumbing, lining and levelling steelwork in accordance with the drawing;
    b. unless otherwise specified, tolerances for the erection of structural steelwork shall be:
        i. levelling ±3 mm (±1/8 in);
        ii. alignment and plumb ±3 mm (±1/8 in).
6. Discrepancies between the erection/shop drawings and the delivered steel members or incorrectly fabricated steel members shall be immediately reported.
7. Temporary erection loads or permanent loads shall not be placed on incomplete portions of the structure being erected, unless approved by the Contractor.
8. Loose timbers, metal sheeting, bolt buckles, tools, debris and temporary scaffolding shall be kept restrained or removed from work areas.
9. The lifting of painted structural members shall be done with a non-abrasive choker.
10. The adequacy and installation of temporary bracing or guy cables required to counteract loadings imposed during erection shall be reviewed and approved by the Contractor.
    a. This responsibility shall also extend to temporary bracing required to ensure safe and stable conditions of partially completed structural assemblies.
11. The top of bearing surfaces and the bottom of base plates shall be cleaned before erection.
    a. Column base plates shall be set and shimmed to the correct positions, elevations and locations as shown on the erection drawings.
    b. Setting nuts, where used, shall be loosened prior grouting.
12. Acceptable methods for measuring bolt pre-tensioning shall be twist-off, “Squirter” Type DTI or turn-of-the-nut method.

13. Anchor bolts shall be snugly tightened as soon as columns are set and grouted.
    a. Anchor bolts shall be fully tightened after the first tier has been plumbed;
    b. For equipment, pre-tensioning shall be as per anchor bolt Manufacturer requirements.
14. All field-cut floor plates or grating openings shall be provided with toe plate protection or banding, as specified on the design drawings.
15. Floor and roof deck sheets shall be installed in accordance with the sheet metal manufacturer’s installation instructions and the design drawings.
16. The procedures and specifications to repair sags, deformations, holes and other irregularities shall be submitted to the Contractor for approval.

6.4 CORRECTION OF ERRORS

1. Fit-up bolts and drift pins shall not be used to bring improperly fabricated members and parts into place (springing).
    a. Drift pins shall not be driven with a force which will damage adjacent metal areas.
2. Standard hole diameters may be enlarged by 1 mm (1/16 in), when necessary, to make connections resulting from minor misfit.
    a. Holes in connections that misfit by more than 1 mm (1/16 in) shall be corrected unless otherwise approved.
3. Enlargement of holes shall be made by reaming or drilling only.
    a. Flame cutting, burning, gouging, chipping or drift punching shall not be permitted.
4. No packing, shimming, filling or wedging shall be permitted to correct faulty work.

7. CONSTRUCTION HSSE REQUIREMENTS

7.1 SAFETY

1. Safety shall be specifically addressed in design, fabrication, erection, inspection and maintenance.
2. The Contractor and/or his sub-contractor (i.e., the Manufacturer/Supplier or Erector) shall have a written safety program that addresses the safety measures for the Manufacturer/Supplier/Fabricator and the Erector to use during fabrication and steel erection work.
3. At a minimum, the safety program shall include:
   a. a detailed description of how to prevent injury to all personnel affected by the fabrication and erection operations;
   b. an effective system for initial orientation and education in safety and accident prevention, as well as appropriate records to document compliance.
4. At a minimum, the field safety program shall place particular emphasis on the following aspects:
   a. SHELL Life Saving Rules;
   b. ground-level preassembly to minimise elevated erection;
   c. hole covers and opening barriers;
   d. access control to incomplete areas of erection;
   e. erection/assembly lifting plans;
   f. hoisting procedures;
   g. safety execution plan;
   h. Task Statement (TS);
   i. Job Safety Analysis (JSA);
   j. Risk Analysis (RA).
5. The safety programs shall be provided to the Contractor for review and comments.

7.2 ERECTION/ASSEMBLY LIFT PLAN

1. The Contractor and/or his sub-contractor (i.e., the Manufacturer/Supplier or Erector) shall submit an assembly lift plan complete with data on heavy lifts, shipping weights and the calculations involved.
2. The Erection/Assembly Lift Plan shall be in compliance with local occupational health, safety, and environmental code requirements.
3. The Contractor and/or his sub-contractor (i.e., the Manufacturer/Supplier or Erector) shall be responsible for assuring that all pre-assemblies that are not specifically shown or noted on the design drawings are preassembled before lifting and will maintain structural integrity during lifting.
4. The Assembly Lift Plan shall demonstrate that the proposed lift will be performed safely, the assemblies being lifted will remain free from distortion or undue bending and will maintain structural integrity during the lift.
5. Access shall be forbidden to personnel not actually engaged in the erection of steelwork, until the structure is finally bolted up, handrails and stair treads are installed and inspected by a qualified field engineer.
   a. Proper green tagging shall indicate if the structure is safe for use.

6. At least one daily contact shall be made with a local weather station, so as to be forewarned for possible high wind.
    a. Sufficient temporary wind bracing shall be used overnight and during weekends and holidays to prevent collapse by wind.
7. At a minimum, the Assembly Lift Plan shall contain the following:
    a. detailed data on the extent of the lifted assembly and its weights;
    b. structural calculations that prove structural stability of the assembled components during lifting operations;
    c. verification of the capacity capabilities for any cranes utilised in the lift;
    d. location and positioning of the cranes;
    e. a description of the rigging to be utilised.
8. Review of the assembly lift plan shall not relieve the Contractor and/or his sub-contractor (i.e., the Manufacturer/Supplier or Erector) of responsibility required for the safe erection and/or lifting of any component, structural assembly or any other item.

7.3 HOLE COVERS AND OPENING BARRIERS

1. Temporary openings in grating floors and stairs shall be clearly blocked by means of scaffolding material or equivalent.
2. Holes shall be covered with a hole-cover conforming to the following:
    a. if one dimension of the opening is 45 cm (18 in) or less, use plywood (multiplex) at least 2 cm (1 in) thick;
    b. if both dimensions of the opening exceed 45 cm (18 in), use two layers of 2 cm (1 in) plywood (multiplex) or material at least 5 cm (2 in) thick;
    c. secure (cleat, wire, nail) all covers to prevent displacement;
    d. clearly mark all covers with a “Danger-hole in walkway - do not remove cover” sign, including name and telephone number of the responsible lead craft in handwriting.

8. QUALITY CONTROL

8.1 GENERAL

1. The Contractor and/or his sub-contractor (i.e., the Manufacturer/Supplier or Erector) shall perform the following:

    a. propose an inspection and test plan for each phase of the design, fabrication and erection of structural steel materials;

    b. establish a system to perform inspections and tests required to assure compliance with the project specifications, local codes and requirements.

8.2 WELDING AND WELD INSPECTION

1. Prior to the commencement of fabrication activities, the Contractor and/or his sub-contractor (i.e., the Manufacturer/Supplier or Erector) shall submit the following documentation for review and approval:

    a. Weld Procedure (WP);

    b. Inspection and Test Procedure (ITP);

    c. Test Procedure Qualifications (TPQ);

    d. Test Acceptance Criteria (TAC);

    e. associated WP specifications covering all joint configurations;

    f. Welder Qualification Records (WQR) and work histories.

2. All welding and WPs shall be in accordance with ANSI/AWS D1.1.

3. Inspectors shall be qualified and certified as ANSI/AWS D1.1 or as per local regulations.

4. Records of all data, tests and examinations relating to WQRs and all WPs used during construction and erection shall be made available for quality review and audits.

5. The design of erection clips for field-welded connections shall be submitted to the Contractor for approval.

6. Run-off bars and extension tabs shall be removed before corrosion protection coatings are applied or before being sent to the field for erection.

7. Field welding, when not specified in drawings, shall be reviewed by the Contractor.

8. Prior approvals from Asset Health Safety Security Environment (HSSE) and necessary work permits shall be obtained for hot works and drilling in a brownfield environment.

8.3 DOCUMENTATION

1. The following documents shall be submitted prior to the start of any fabrication and construction activities:
   a. Inspection and Test Plan (ITP);
   b. inspection procedure document that shall provide details of how compliance with the requirements in this GFS;
      i. the design drawings, shop and erection drawings shall be achieved;
   c. Welding Procedure Specification (WPS);
   d. Welder(s) and Welder Inspector(s) qualification and certification records;
   e. quality control program, quality plan, quality control procedure, including Quality Control Records (QCRs), records of quality control inspection test reports and test results;
   f. Procedure Qualification Record;
   g. Material Handling Procedure;
   h. Materials and Workmanship Specifications;
   i. Mill Test Certificates;
   j. Material Certificates for bolts and nuts;
   k. Test Acceptance Criteria;
   l. list of welders;
   m. temporary works - calculations and drawings;
   n. one set of reproducible final erection and shop drawings;
   o. maintain a complete to date (As-Built) set of erection drawings at the job site;
   p. method statements covering ‘Erection and Construction Methods’, including temporary facilities (including scaffolding) and HSSE;
   q. correction of non-conformances including repair procedures.
2. The Contractor may reject improper, inferior, defective or unsuitable materials, documentation and workmanship.
3. All materials, installations and workmanship that is rejected shall be repaired or replaced by the Contractor and/or his sub-contractor (i.e., the Manufacturer/Supplier or Erector).

9. REFERENCES

In this GFS, reference is made to the following publications:

NOTES:
1. Unless specifically designated by date, the latest edition of each publication shall be used, together with any amendments/supplements/revisions thereto.
2. The DEPs, SPEPs, GFSs and most referenced external standards are available to Shell staff on the SWW (Shell Wide Web) at http://sww.shell.com/standards/.

SHELL STANDARDS

Onshore Loading Specification for Onshore LNG Facilities	GFS IG.34.85.02-Gen.
Painting and Coating Specification for Onshore LNG Facilities	GFS IG.31.87.77-Gen.
Heavy Machinery Foundations for Onshore LNG Facilities	GFS IG.34.85.25-Gen.

AMERICAN STANDARDS

Specification for Structural Steel Buildings	ANSI/AISC 360
Code of Standard Practice for Steel Buildings and Bridges	ANSI/AISC 303
AISC Seismic Design Manual	ANSI/AISC 327
Steel Construction Manual	ANSI/AISC 325
Seismic Provisions for Structural Steel Buildings	ANSI/AISC 341
Prequalified Connections for Special and Intermediate Steel Moment Frames for Seismic Applications	ANSI/AISC 358
Structural Welding Code-Steel	ANSI/AWS D1.1
Standard Specification for Carbon Structural Steel	ASTM A36/A36M
Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless	ASTM A53/A53M
Standard Specification for Seamless Carbon Steel Pipe for High-Temperature Service	ASTM A106/A106M
Standard Specification for Cold-Formed Welded and Seamless Carbon Steel Structural Tubing in Rounds and Shapes	ASTM A500/A500M
Standard Specification for Hot-Formed Welded and Seamless Carbon Steel Structural Tubing	ASTM A501
Standard Specification for Structural Steel Shapes	ASTM A992/A992M
Standard Specification for Steel, Sheet and Strip, Hot-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, and Ultra-High Strength	ASTM A1011/A1011M
Standard Specification for Hot-Rolled Carbon, Low-Alloy, High-Strength Low-Alloy, and Alloy Steel Floor Plates	ASTM A786/A786M
Standard Specification for Carbon Steel Bolts and Studs, 60,000 PSI Tensile Strength	ASTM A307
Standard Specification for Structural Bolts, Steel, Heat Treated, 120/105 ksi Minimum Tensile Strength	ASTM A325/A325M
Standard Specification for Carbons and Alloy Steel Nuts	ASTM A563
Standard Specification for Hardened Steel Washers	ASTM F436/F436M
Standard Specification for Compressible Washer Type Direct Tension Indicators for Use with Structural Fasteners	ASTM F959

Standard Specification for High-Strength Steel Bolts, Classes 10.9 and 10.9.3, for Structural Steel Joints (Metric)	ASTM A490
Standard Specification for Structural Bolts, Alloy Steel, Heat Treated, 150 ksi Minimum Tensile Strength	ASTM A490/A490M
Standard Specification for Zinc (Hot-Dip Galvanised) Coatings on Iron and Steel Products	ASTM A123/A123M
Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware;	ASTM A153
Steel Construction Manual	AISC SCM
Metal Bar Grating Manual	ANSI/NAAMM MBG 531

PROJECT STANDARDS

The project and Engineering Contractor shall apply the standards as specified above and no project-specific technical standards and/or specifications shall be made on top of these.

It is recognized that the Engineering Contractor has its own procedures and standards to manage a project, from that perspective it is allowed to develop a limited number of project specific standards, specifications, procedures that are required to manage and administer the project execution.